

SIJIA LIAO

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EDUCATION

Doctor of Philosophy, University of Arizona

Statistics and Data Science, GPA: 4.0/4.0

Aug. 2020 – May 2026

Tucson, Arizona, USA

- Dissertation: *Sparse and Smooth Estimation for High-Dimensional Structured Regression*
- Advisors: Hao Helen Zhang and Xiaoxiao Sun
- **Areas:** Machine Learning, Statistical Learning, High-Dimensional Statistics, Functional Data Analysis, Optimization

Master of Science, Sun Yat-Sen University

Probability and Mathematical Statistics, GPA: 4.0/4.0

Aug. 2017 – Jul. 2020

Guangzhou, Guangdong, China

Bachelor of Science, Sun Yat-Sen University

Mathematics and Applied Mathematics, GPA: 3.6/4.0

Aug. 2013 – Jul. 2017

Guangzhou, Guangdong, China

EXPERIENCE

Research Scientist Intern

TikTok Inc.

Jun. 2025 – Dec. 2025

Bellevue, Washington, USA

- Developed a rationale-augmented vision model training framework for video content understanding, improving classification performance by 2–5% F1.
- Designed a dynamic EMA checkpoint ensembling algorithm adopted in the training pipeline, yielding an additional 2–4% F1 improvement and more stable performance.
- Conducted hard-case mining and error analysis to identify model failure modes and guide iterative model improvements.

Graduate Research Assistant/Associate

Department of Epidemiology and Biostatistics

Jan. 2023 – May 2025

Tucson, Arizona, USA

- Developed a high-dimensional nonparametric learning framework that jointly performs variable selection and smoothing, supported by theoretical guarantees and extensive simulations.
- Developed an interpretable scalar-on-image regression method that identifies sparse and spatially smooth predictive regions in imaging data.
- Achieved 10% lower prediction error than competing approaches while improving identification of non-predictive regions by over 90%.

Graduate Teaching Assistant

Department of Mathematics, University of Arizona

Aug. 2021 – Dec. 2024

Tucson, Arizona, USA

- Teaching assistant for graduate Theory of Probability, Introduction to Statistics and Biostatistics, and College Algebra.

SELECTED PUBLICATIONS

Liao, S., Sun, X., Hao, N., and Zhang, H. H. *Interpretable Scalar-on-Image Linear Regression Models via the Generalized Dantzig Selector*. **Accepted for publication in the *Journal of the American Statistical Association (JASA)***, 2026. [[arXiv](#)] [[GitHub](#)]

Liao, S. and Zhang, H. H. *Component Selection and Smoothing in High-Dimensional Additive Models*. Manuscript in preparation for submission.

HONORS & AWARDS

ASA Nonparametric Statistics Section Student Paper Award, Finalist, 2025. Invited presentation at Joint Statistical Meetings (JSM) 2025. [[Abstract](#)]

Herbert E. Carter Travel Award, University of Arizona Graduate College, 2025.

Outstanding Graduate Teaching Assistant Award, Honorable Mention, University of Arizona, 2024.